

JEDEC STANDARD

Temperature, Bias, and Operating Life

JESD22-A108G

(Revision of JESD22-A108F dated July 2017)

NOVEMBER 2022

JEDEC SOLID STATE TECHNOLOGY ASSOCIATION



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TEMPERATURE, BIAS, AND OPERATING LIFE**Contents**

	Page
1 Scope	1
2 Apparatus	1
2.1 Circuitry	1
2.1.1 Device Schematic	1
2.1.2 Power	1
2.2 Device Mounting	1
2.3 Power Supplies and Signal Sources	2
2.4 Environmental Chamber	2
3 Definitions.....	2
3.1 Maximum Operating Voltage	2
3.2 Absolute Maximum Rated Voltage	2
3.3 Absolute Maximum Rated Junction Temperature	2
4 Procedure.....	2
4.1 Stress Duration.....	2
4.2 Stress Conditions	3
4.2.1 Ambient Temperature	3
4.2.2 Operating Voltage.....	3
4.2.3 Biasing Configurations	3
4.2.3.1 High Temperature Forward Bias (HTFB).....	3
4.2.3.2 High Temperature Operating Life (HTOL) / Low Temperature Operating Life (LTOL)	3
4.2.3.3 High Temperature Reverse Bias (HTRB)	4
4.2.3.4 High Temperature Gate Bias (HTGB)	4
5 Cool-Down	4
6 Measurements	4
7 Failure Criteria	5
8 Life Test Reporting.....	5
Annex A (Informative) Differences Between JESD22-A108F and Its Predecessors	6

Tables**Page**

Table 1 — Additional Stress Requirements for Parts Not Tested Within Time Window	5
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TEMPERATURE, BIAS, AND OPERATING LIFE

(From JEDEC Board Ballot JCB-22-55, formulated under the cognizance of the JC-14.1 Committee on Reliability Test Methods for Packaged Devices.)

1 Scope

This test is used to determine the effects of bias conditions and temperature on solid state devices over time. It simulates the devices' operating condition in an accelerated way, and is primarily for device qualification and reliability monitoring. A form of high temperature bias life using a short duration, popularly known as burn-in, may be used to screen for infant mortality-related failures. The detailed use and application of burn-in is outside the scope of this document.

1.1 Applicable Documents

JESD47, *Stress-Test Driven Qualification of Integrated Circuits*

JEP122, *Failure Mechanism and Models for Silicon Semiconductor Devices*

2 Apparatus

The performance of this test requires equipment that is capable of providing the particular stress conditions to which the test samples will be subjected.

2.1 Circuitry

The circuitry through which the samples will be biased must be designed with several considerations:

2.1.1 Device Schematic

The biasing and operating schemes must consider the limitations of the device and shall not overstress the devices or contribute to thermal runaway.

2.1.2 Power

The test circuit should be designed to limit power dissipation such that, if a device failure occurs, excessive power will not be applied to other devices in the sample.

2.2 Device Mounting

Equipment design, if required, shall provide for mounting of devices to minimize adverse effects while parts are under stress, (e.g., improper heat dissipation).

2.3 Power Supplies and Signal Sources

Instruments (such as DVMs, oscilloscopes, etc.) used to set up and monitor power supplies and signal sources shall be calibrated and have good long-term stability.

2.4 Environmental Chamber

The environmental chamber shall be capable of maintaining the specified temperature within a tolerance of $\pm 5^{\circ}\text{C}$ throughout the chamber while parts are loaded and unpowered.

3 Definitions

3.1 Maximum Operating Voltage

The maximum supply voltage at which a device is specified to operate in compliance with the applicable device specification or data sheet.

3.2 Absolute Maximum Rated Voltage

The maximum voltage that may be applied to a device, beyond which damage (latent or otherwise) may occur; it is frequently specified by device manufacturers for a specific device and/or technology.

3.3 Absolute Maximum Rated Junction Temperature

The maximum junction temperature of an operating device, beyond which damage (latent or otherwise) may occur; it is frequently specified by device manufacturers for a specific device and/or technology.

NOTE Manufacturers may also specify maximum case temperatures for specific packages.

4 Procedure

The sample devices shall be subjected to the specified or selected stress conditions for the time and temperature required.

4.1 Stress Duration

The bias life duration is intended to meet or exceed an equivalent field lifetime under application use conditions. The duration is established based on the acceleration of the stress(see JEP122). The stress duration is specified by application qualification requirements, JESD47 or the applicable procurement document. Interim measurements may be performed as necessary per restrictions in clause 6.

4.2 Stress Conditions

The stress condition shall be applied continuously (except during interim measurement periods). The time spent elevating the chamber to accelerated conditions, reducing chamber conditions to room ambient, and conducting the interim measurements shall not be considered a portion of the total specified test duration.

4.2.1 Ambient Temperature

The ambient temperature and bias for high temperature stress shall be adjusted to result in a minimum junction temperature of the devices under stress of 125°C unless otherwise specified as for extended use or other environments. Unless otherwise specified, the ambient temperature for low temperature stress shall be a maximum of –10°C. For products which experience temperature variations during high temperature stress, it is acceptable to use higher or lower junction temperatures for specified blocks/dynamic stress patterns as long as application lifetime equivalent stress duration targets are achieved.

NOTE Devices designed for use in an extended temperature environment may be stressed at temperatures which may extend up to 250°C. The stress temperature may exceed the operating temperature but not the absolute maximum rated temperature and voltage of the technology.

4.2.2 Operating Voltage

Unless otherwise specified, the operating voltage should be the maximum operating voltage specified for the device. When the maximum operating voltage is not used, report the voltage used during the qualification. A higher voltage is permitted in order to obtain lifetime acceleration from voltage as well as temperature; this voltage must not exceed the absolute maximum rated voltage for the device, and must be agreed upon by the device manufacturer.

4.2.3 Biasing Configurations

Biasing configurations may be bias stress (static or pulsed) or operating stress (dynamic). Depending upon the biasing configuration, supply and input voltages may be grounded or raised to a maximum potential chosen to ensure a stressing temperature not higher than the maximum-rated junction temperature. Device outputs may be unloaded or loaded, to achieve the specified output voltage level. If a device has a thermal shutdown feature it shall not be biased in a manner that could cause the device to go into thermal shutdown.

4.2.3.1 High Temperature Forward Bias (HTFB)

The HTFB test is configured to forward bias major power handling junctions of the device samples. The devices may be operated in either a static or a pulsed forward bias mode. Pulsed operation is used to stress the devices at, or near, maximum-rated current levels. The particular bias conditions should be determined to bias the maximum number of the solid state junctions in the device. The HTFB test is typically applied on power devices, diodes, and discrete transistor devices (not typically applied to integrated circuits).

4.2.3.2 High Temperature Operating Life (HTOL) / Low Temperature Operating Life (LTOL)

The HTOL / LTOL test is configured to bias the operating nodes of the device samples. The devices may be operated in a dynamic operating mode. Typically, several input parameters may be adjusted to control internal power dissipation. These include: supply voltages, clock frequencies, input signals, etc., that may be operated even outside their specified values, but resulting in predictable and nondestructive behavior of the devices under stress.

4.2.3.2 HTOL / LTOL (cont'd)

The particular bias conditions should be determined to bias the maximum number of potential operating nodes in the device. The HTOL test is typically applied on logic and memory devices. The LTOL test is intended to look for failures caused by hot carriers, and is typically applied on memory devices or devices with submicron device dimensions.

4.2.3.3 High Temperature Reverse Bias (HTRB)

The HTRB test is configured to reverse bias major power handling junctions of the device samples. The devices are characteristically operated in a static operating mode at, or near (typically 80% to 100%) of maximum-rated breakdown voltage and/or current levels. The particular bias conditions should be determined to bias the maximum number of the solid state junctions in the device. The HTRB test is typically applied on power devices

4.2.3.4 High Temperature Gate Bias (HTGB)

The HTGB test biases gate or other oxides of the device samples. The devices are normally operated in a static mode at, or near, maximum-rated oxide breakdown voltage levels. The particular bias conditions should be determined to bias the maximum number of gates in the device. The HTGB test is typically used for power devices.

5 Cool-Down

Devices on high temperature stress shall be cooled to 55°C or lower before removing the bias. Cooling under bias is not required for a given technology if verification data is provided by the manufacturer. The interruption of bias for up to one minute, for the purpose of moving the devices to cool-down positions separate from the chamber within which life testing was performed, shall not be considered removal of bias. All specified electrical measurements shall be completed prior to any reheating of the devices, except for interim measurements subject to restrictions of clause 6.

NOTE Bias refers to application of voltage to power pins.

6 Measurements

The measurements specified in the applicable life test specification shall be made initially, at the end of each interim period, and at the conclusion of the life test. Interim and final measurements may include high temperature testing. However, testing at elevated temperatures shall only be performed after completion of specified room (and lower) temperature test measurements. After interim testing, bias shall be applied to the parts before heat is applied to the chamber, or within ten minutes of loading the final parts into a hot chamber. Electrical testing shall be completed as soon as possible after removal of bias from devices and no longer than 96 hours for high voltage devices (defined as > 10 volts) or 168 hours for all other devices. If the availability of test equipment or other factors make meeting this requirement difficult, bias must be maintained on the devices; this may be either at the stress temperature or room temperature and the bias may be reduced to nominal voltage from any accelerated voltage in use for stress. If the devices have been removed from bias and the time window is exceeded, the stress must be resumed for the duration specified in Table 1 prior to completion of the measurements.

6 Measurements (cont'd)

After an interim measurement, the stress shall be continued from the point of interruption. The time window and the high temperature testing restrictions of this clause need not be met if verification data for a given technology is provided.

Table 1 — Additional Stress Requirements for Parts Not Tested Within Time Window

Hours by which time window has been exceeded	> 0 but ≤ 168	> 168 but ≤ 336	> 336 but ≤ 504	Other
Additional stress hours required prior to performing electrical test	The lesser of 24 OR 50% of the stress interval since the last test readout	The lesser of 48 OR 50% of the stress interval since the last test readout	The lesser of 72 OR 50% of the stress interval since the last test readout	The lesser of 24 hours for each 168 hours (week) by which the time window has been exceeded OR 50% of the stress interval since the last test readout

7 Failure Criteria

A device is defined as a failure if it does not meet the requirements of the applicable procurement document.

8 Life Test Reporting

The following items shall be specified in the applicable life test specification:

- Special preconditioning, when applicable.
- Stress temperature (chamber ambient).
- Stress duration.
- Stress mounting, if special instructions are needed.
- Stress condition and stress circuit schematic.
- Sample size and acceptance number.
- Time to complete endpoint measurements, if other than specified in clause 6.
- Operating mode.
- Interim read points, if required.
- Maximum junction temperature during stress.
- Verification data if cool-down under bias is not performed.

Annex A (Informative) Differences Between JESD22-A108F and Its Predecessors

These sections briefly describe most of the differences between the text of this standard, JESD22-A108G, and its predecessors JESD22-A108F, JESD22-A108E, JESD22-A108D (November 2010), JESD22-A108C (June 2005), and JESD22-A108B (December 2000).

Difference between JESD22-A108G and JESD22-A108F

Clause	Description of change
4.2.2	Rephrased the paragraph to allow lower operating voltages under any condition. Added: When the maximum operating voltage is not used, report the voltage used during qualification
4.2.3.3	Added the phrase (typically 80% to 100%) to the sentence
8	Changed title from Summary to Life Test Reporting to clarify the section purpose

Difference between JESD22-A108F and JESD22-A108E

Clause	Description of change
4.1	Reword to cite application-based use conditions and qualification requirements
4.2.1	Acknowledge provision for possible stress needs above 125°C in body and with added Note and for products which experience temperature variations during high temperature stress

Differences between JESD22-A108E and JESD22-A108D

Clause	Description of change
6	Clarified allowance to remain on bias at non-accelerated conditions and revised restress requirements.
6	Capped restress requirements for short stress intervals.

Differences between JESD22-A108D and JESD22-A108C

Clause	Description of change
6	Revised time window requirements in clause 6.
6	Revised table 1 to align with clause 6.

Differences between JESD22-A108C and JESD22-A108-B

Clause	Description of change
6	Revised wording in end of clause 6 and replaced note with table 1.
8	Corrected 8(g) to refer to clause 6 instead of paragraph 5.



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The purpose of this form is to provide the Technical Committees of JEDEC with input from the industry regarding usage of the subject standard. Individuals or companies are invited to submit comments to JEDEC. All comments will be collected and dispersed to the appropriate committee(s).

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1. I recommend changes to the following:

☐ Requirement, clause number _____

☐ Test method number _____ Clause number _____

The referenced clause number has proven to be:

☐ Unclear ☐ Too Rigid ☐ In Error

☐ Other _____

-
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-
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